Homework 7

MAT 2040 : Summer 2023 : Prof. Lun Yu

Problem 1

Find the eigenvalues and the corresponding eigenspaces for each of the following matrices:

1. $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$
2. $\begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
4. $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
5. $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{pmatrix}$

Problem 2

Let A be an $n \times n$ matrix. Prove that A is singular if and only if $\lambda = 0$ is an eigenvalue of A.

Problem 3

1. Let A be a nonsingular matrix and let λ be an eigenvalue of A. Show that $1/\lambda$ is an eigenvalue of A^{-1}.

Problem 4

1. An $n \times n$ matrix A is said to be idempotent if $A^2 = A$. Show that if λ is an eigenvalue of an idempotent matrix, then λ must be either 0 or 1.

Problem 5

1. Let A be an $n \times n$ matrix and let $B = A + I$. Is it possible for A and B to be similar? Explain.

Problem 6

1. Let Q be an orthogonal matrix. Show that if λ is an eigenvalue of Q, then $|\lambda| = 1$.
2. Show that $|\det(Q)| = 1$.

Problem 7

1. In each of the following, factor the matrix A into a product XDX^{-1}, where D is diagonal:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2.

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

Problem 8

1. For each of the matrices in the previous exercise, use the XDX^{-1} factorization to compute A^6.

Problem 9

Let A be a diagonalizable matrix whose eigenvalues are all either 1 or -1 . Show that $A^{-1} = A$.

Problem 10

Show that any 3×3 matrix of the form $\begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & b \end{bmatrix}$ is defective.

Problem 11

For each of the following, find all possible values of the scalar α that make the matrix defective or show that no such values exist.
(a)

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & \alpha \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 3\alpha & 1 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix}$$

Problem 12

Let A be a diagonalizable $n \times n$ matrix. Prove that if B is any matrix that is similar to A, then B is diagonalizable.

Problem 13

Show that if A and B are two $n \times n$ matrices with the same diagonalizing matrix X, then $AB = BA$.

Problem 14

For each of the following pairs of vectors $\mathbf{z}$ and $\mathbf{w}$, compute (i) $\|\mathbf{z}\|$, (ii) $\|\mathbf{w}\|$, (iii) $\langle \mathbf{z}, \mathbf{w} \rangle$, and (iv) $\langle \mathbf{w}, \mathbf{z} \rangle$:

(a)

$$\mathbf{z} = \begin{bmatrix} 1+2i \\ 3i \end{bmatrix}, \mathbf{w} = \begin{bmatrix} -5 \\ 3+i \end{bmatrix}$$

(b)

$$\mathbf{z} = \begin{bmatrix} 1+4i \\ 2i \\ 3-2i \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 2-i \\ 4 \\ 2i \end{bmatrix}$$

Problem 15

Let $\{\mathbf{u}_1, \mathbf{u}_2\}$ be an orthonormal basis for $\mathbb{R}^2$, and let $\mathbf{z} = 4\mathbf{u}_1 + 5\mathbf{u}_2$.

(a) What are the values of $\mathbf{u}_1^T\mathbf{z}, \mathbf{z}^T\mathbf{u}_1, \mathbf{u}_2^T\mathbf{z}$ and $\mathbf{z}^T\mathbf{u}_2$?

(b) Determine the value of $\|\mathbf{z}\|$.

Problem 16

Which of the matrices that follow are Symmetric? Orthogonal?

(a)

$$\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$$

(b)

$$\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{\sqrt{3}} \\ \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 17

Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^n$ and define $U = I - 2\mathbf{u}\mathbf{u}^T$. Show that U is both orthogonal and symmetry and, consequently, is its own inverse.

Problem 18

find the singular value decomposition of each of the following matrices:
(a)

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Problem 19

Find the matrix associated with each of the following quadratic forms:
(a) $3x^2 - 5xy + y^2$
(b) $2x^2 + 3y^2 + z^2 + xy - 2xz + 3yz$

Problem 20

Show that if A is a symmetric positive definite matrix, then A is nonsingular and A^{-1} is also positive definite.